

**STATE OF CALIFORNIA
CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL COAST REGION
895 Aerovista, Suite 101
San Luis Obispo, California 93401**

**WASTE DISCHARGE AND WATER RECYCLING REQUIREMENTS
ORDER NO. R3-2004-0153
Waste Discharger Identification No. 3359803001**

**For
SAN BENITO COUNTY
COUNTY SERVICES AREA NO. 45
RANCHO LARIOS
San Benito County**

The California Regional Water Quality Control Board, Central Coast Region, (hereafter the Board) finds:

1. Mr. Rene Fuog submitted a complete Report of Waste Discharge on June 27, 2004, in accordance with Section 13260 of the California Water Code. The report was filed on behalf of the County of San Benito (hereafter Discharger) for authorization to discharge recycled water within the South Santa Clara Valley Hydrologic Area. Groundwater basins are known locally as Hollister and San Juan Valley basins.
2. The Discharger operates a tertiary wastewater treatment plant and spray irrigation area for treatment and reuse of wastewater from the subdivision. The facility is owned by the Larwin Company, operated by Bracewell Engineering under a contract issued by the San Benito County Public Works and paid for by the residents of Rancho Larios (County Service Area No. 45).
3. Recycled water reuse is proposed as the primary method of disposal for Rancho Larios subdivision. The subdivision is shown on attachments "A" and "B" of this Order and will include up to 153 residential lots, a recreation facility, and a possible equestrian facility at build out. The facility is located on 702 acres west of the City of San Juan Bautista in San Benito County.
4. Average dry weather flow from the subdivision is estimated at 38,000 gallons per day. Estimated flow is based on per household water consumption rates for similar residences in the area. Secondary treatment consists of a sequential batch reactor with screening, grit removal, oxidation, standby power and system monitoring and alarms and flow equalization. Final filtration and chlorine disinfection will provide the tertiary level of treatment.
5. The Discharger currently uses spray irrigation as the primary method of waste disposal. The spray irrigation area is designated agricultural open space for grazing. Recycled water irrigates fodder, fiber, and seed crops. Title 22 reclamation will occur on soccer fields and landscaping within the development as identified on Attachment B.
6. Soils beneath the spray disposal and reclamation areas consist of silty sandstone. Depth to first encountered groundwater is from 20 to 30 feet below ground surface.

7. Recycled water is stored in a lined reservoir. The reservoir has a storage capacity of 15.8 acre-feet and provides storage of effluent for up to 122 days at the average design flow rate.
8. State Department of Health Services' Department criteria for use of recycled water is in Title 22, Chapter 3, of the California Code of Regulations. The Board has consulted with this Department regarding the regulation of this discharge.
9. Requirements specified in this Order ensure proper use of recycled water for the protection of public health. Recycled water treatment and spray irrigation in compliance with these requirements will not pose a significant threat to underlying groundwater resources.
10. These waste discharge requirements are for an existing facility and are exempt from the provisions of the California Environmental Quality Act (Public Resources Code, Section 21000, et. seq.) in accordance with Section 15301, Article 19, Chapter 3, Division 6, Title 14 of the California Code of Regulations.
11. Discharge of waste is a privilege, not a right, and authorization to discharge is conditional upon the discharge complying with provisions of Division 7 of the California Water Code and any more stringent effluent limitations necessary to implement water quality control plans, to protect beneficial uses, and to prevent nuisance. Compliance with this Order should ensure this and mitigate any potential adverse changes in water quality due to the discharge.
12. On September 17, 2004, the Board notified the Discharger and interested persons of its intent to issue waste discharge requirements, provided them with an opportunity to submit their written views and recommendations, and scheduled a public hearing.
13. In a public hearing December 3, 2004, the Board heard and considered all comments pertaining to the discharge and found this Order consistent with the above findings.
14. The Water Quality Control Plan, Central Coastal Basin, (Basin Plan) was adopted by the Board on November 17, 1989, and approved by the State Board on August 16, 1990. The Board approved amendments to the Basin Plan on February 11, 1994 and September 8, 1994. The Basin Plan incorporates statewide plans and policies by reference and contains a strategy for protecting beneficial uses of State waters.
15. Present and anticipated beneficial uses of groundwater in the vicinity of the discharge include:
 - a) Domestic Supply,
 - b) Industrial Supply, and
 - c) Agricultural Supply.
16. Groundwater objectives for the Hollister Sub-basin are as follows:

Table 1: Median Groundwater Objectives for the Pajaro River sub-basin

Parameter (mg/L)	Sub-area Hollister
TDS	1200
Cl	150
Sulfate	250
Sodium	200
Nitrate as N	5
17. The San Benito River is the closest surface water body to the Rancho Larios development although impacts from this discharge would be highly unlikely. The Basin Plan designates the following existing and anticipated beneficial uses of the San Benito River that could be affected by the discharge:
 - a) Municipal and Domestic Supply;
 - b) Agricultural Water Supply;
 - c) Industrial Service Supply;
 - d) Groundwater Recharge;
 - e) Water Contact Recreation;
 - f) Non-Contact Water Recreation;
 - g) Wildlife Habitat;
 - h) Warm Freshwater Habitat;

- i) Spawning, Reproduction, and/or Early Development;
 - j) Freshwater Replenishment;
 - k) Commercial and Sport Fishing.
18. For receiving waters with designated beneficial uses of municipal and domestic water supply, the Basin Plan establishes the primary drinking water maximum contaminant levels (MCLs), listed at Title 22 of the California Code of Regulations, Sections 64431 (inorganic compounds) and 64444 (organic compounds), as applicable water quality objectives.
19. The Basin Plan specifies water quality objectives for certain surface waters, which are intended to serve as a baseline for evaluating water quality management in the basin. The objectives are, at best, representative of gross areas only, and are based on preservation of existing quality or water quality enhancement believed attainable following control of point sources. Water quality objectives are as follows for the San Benito River sub-area of the Parajo River sub-basin.

Table 2: Surface Water Quality Objectives for the Pajaro River sub-basin

Parameter (mg/L)	Sub-area San Benito River
Total Dissolved Solids	1400
Chloride	200
Sulfate	350
Boron	1.0
Sodium	250

Excerpted from Table 3-7, page III-13 of the Basin Plan

20. Monitoring and Reporting Program (MRP) No. R3-2004-0153 is a part of the proposed Order. The MRP requires routine monitoring of water supply, influent, secondary treatment process effluent, tertiary treatment process effluent, disposal areas, and recycled water use areas to verify compliance with this order and protection of water quality.

IT IS HEREBY ORDERED, pursuant to authority in Sections 13263 and 13523 of

the California Water Code, that San Benito County Service Area No. 45, its agents, successors, and assigns, may spray irrigate or reclaim recycled water providing they comply with the following:

[Requirements are based on specific regulations, Regional Board staff's professional judgment, and recommendations from state and county environmental health agencies for the sufficient protection of public health and the environment. Specific regulations include the Basin Plan (superscript "1") and Title 22 (superscript "2") requirements.]

All technical and monitoring reports submitted pursuant to this Order are required pursuant to Sections 13267 of the California Water Code. Failure to submit reports in accordance with schedules established by this Order, attachments to this Order, or failure to submit a report of sufficient technical quality to be acceptable to the Executive Officer, may subject the Discharger to enforcement action pursuant to Sections 13268 of the California Water Code. The Regional Board will base all enforcement actions on the date of Order adoption.

A. PROHIBITIONS

1. Discharge of recycled water to areas other than the spray irrigation areas shown in "Attachment B" is prohibited. Any water not meeting the Title 22 tertiary 2.2 requirements must be disposed in the existing fenced open space irrigation area.
2. Discharge of recycled water to adjacent drainage ways and adjacent properties is prohibited.
3. Daily flow averaged over each month greater than 38,000 gallons is prohibited.
4. Introduction of any substance into the waste stream that adversely affects the collection, treatment, or disposal facility is prohibited.
5. Storage pond freeboard of less than 2 feet is prohibited.

6. Bypass of treatment facilities and discharge of untreated or partially treated waste is prohibited.

B. RECLAMATION SPECIFICATIONS

1. Reclaimed water, as sampled in plant effluent, shall not contain constituents in excess of the following:

Parameter	Monthly Average (mg/l)
BOD ₅	30
Settleable Solids	0.5 ml/l
TDS	1,200
Cl	350
Sodium	250
Nitrate as N	10

2. Reclaimed water delivered shall not have a pH less than 6.5 or greater than 8.4¹.

C. TERTIARY-2.2 EFFLUENT LIMITATIONS

1. All tertiary 2.2 recycled water produced shall conform with recycled water criteria contained in Title 22, Division 4, Chapter 3, of the California Code of Regulations.
2. Tertiary treatment process effluent turbidity shall not exceed an average of 2 NTU within a 24-hour period; 5 NTU more than 5 percent of the time within a 24-hour period; or 10 NTU at any time.
3. The median concentration of total coliform bacteria measured in the disinfected effluent shall not exceed 2.2 MPN per 100 ml, utilizing the bacteriological results of the last seven days for which analyses have been completed. Total coliform bacteria shall not exceed 23 MPN per 100 ml in more than one sample in any 30-day period. No total coliform bacteria sample shall exceed 240 MPN per 100 ml.
4. Recycled water shall be confined within the designated recycled water use area

shown in Attachment B and labeled as soccer field, tennis courts, and volleyball lawn area. Excess can be discharged to the existing open space irrigation area adjacent to the storage pond.

5. Drinking water fountains shall be protected against contact with recycled water spray, mist, or runoff.
6. Recycled water spray, mist, or runoff shall not enter dwellings, designated outdoor eating areas, or food handling facilities.
7. Valves in the recycled water irrigation system shall be designed and constructed so unauthorized persons cannot open them.
8. Proper backflow and cross-connection protection for domestic water services and irrigation wells shall be provided.
9. Hose bibs or other types of hose connections installed in the recycled water irrigation system shall be of different sizes or have other measures incorporated to preclude interchange of hoses between fresh and recycled water-irrigation systems.
10. Cross connections between potable water supply and recycled water system are prohibited as stipulated in Section 60310(g) of the Water Recycling Code. The recycled water system shall not include hose bibs, pursuant to Section 60310(h) of the Water Recycling Code.

D. SECONDARY-23 USE SPECIFICATIONS

1. Water that fails to meet the above turbidity and coliform limits for tertiary-2.2 reclaimed water, but still does not exceed a Most Probable Number (MPN) of 23 per 100 may be used for limited reclamation purposes such as dust control or soil compaction in accordance with the following requirements:
 - a) User shall instruct, orally and in writing, each recycled water tank truck driver as to

the requirements of this Order and the potential health hazards involved with the use of recycled water.

- b) Recycled water shall be applied so as to minimize aerosol formation during spraying.
- c) Recycled water shall be applied so as to prevent public or employee contact with the water.
- d) The User shall ensure that recycled water use areas are posted with signs informing the site personnel that recycled water is in use.
- e) All trucks used to haul recycled water must have water tight valves and fittings, must not leak, must be clean of other contaminants before being used to haul recycled water, cannot be used to haul water for potable uses, and cannot be connected to potable water supply systems.
- f) Recycled water systems shall be properly labeled and regularly inspected to ensure proper operation, absence of leaks, and absence of illegal connections.

E. GENERAL SPECIFICATIONS FOR RECLAMATION AND REUSE

All reclamation activities must be consistent with the following general requirements:

- 1. Recycled water shall not be used for irrigation during periods of extended rainfall or runoff.
- 2. Recycled water shall not be applied, or impounded, within 50 feet of any domestic water supply well.
- 3. Personnel involved in producing, transporting, or using recycled water shall be informed of possible health hazards that may result from contact and use of recycled water.
- 4. Use of recycled water shall occur at a time and in a manner to prevent or minimize public contact with effluent.

- 5. Discharger shall operate the recycled water system to minimize ponding or puddling in the irrigated area.
- 6. Discharger's employees shall be informed of possible hazards associated with contact or use of recycled water.
- 7. All recycled water reservoirs and other areas with public access shall be posted (in English and Spanish) with signs that are visible to the public, in a size no less than 4 inches high by 8 inches wide, that include the following wording: "RECYCLED WATER - DO NOT DRINK."
- 8. The incidental discharge of recycled water to waters of the State is not a violation of these requirements if the incidental discharge does not unreasonably affect the beneficial uses of the water, and does not result in exceedances of an applicable water quality objective in the receiving water.
- 9. Perimeter warning signs for the spray irrigation area shall be posted every 1250 feet. At a minimum, signs shall be posted at each corner and at access roads. Signs shall provide information consistent with Department of Health guidelines Section 60310(f).
- 10. Recycled water pipes shall be colored purple or wrapped in purple color tape.
- 11. A registered engineer shall inspect the treatment system upon completion of any modifications to the treatment plant to ensure conformance with design plans. The Discharger shall submit an engineering report, including as built drawings, to the Board within 30 days of system modification certifying proper installation and operation.

F. DISPOSAL

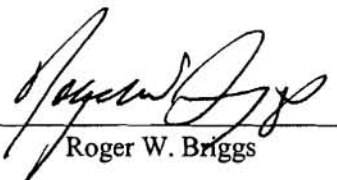
- 1. Effluent that does not meet either of the above reclamation criteria must be discharged to the existing fenced disposal

field.

G. PROVISIONS

1. Discharger shall comply with "Monitoring and Reporting Program No. R3-2004-0153," as specified by the Executive Officer.
2. Discharger shall comply with all items of the attached "Standard Provisions and Reporting Requirements for Waste Discharge Requirements," dated January, 1984 (also referred to as "Standard Provisions"), except Item Nos. A.8 and C.16.
3. The Discharger shall make every reasonable effort to reduce the salts loading caused by the reclamation of this wastewater, including the encouragement of cartridge style over regenerating water softeners. The discharger shall submit for Executive Officer approval a salts management plan by **December 3, 2005**. The purpose of this plan will be to both accurately identify the actual land carrying capacity for salt discharge, and to identify and plan the implementation of all reasonable means for achieving this loading. In addition to this plan, updates will be submitted with the annual monitoring report (starting in 2006) to document the progress towards this goal.
4. Order No. 98-75, "Waste Discharge and Recycled Water Requirements San Benito County Service Area No. 45, Rancho Larios" adopted by the Board on October 23, 1998, is hereby rescinded.
5. Pursuant to Title 23, Division 3, Chapter 9, of the California Code of Regulations, the Discharger must submit a written report to the Executive Officer not later than July 1, 2015, addressing:
 - a. Whether there will be changes in the continuity, character, location, or volume of the discharge; and,
 - b. Whether, in their opinion, there is any portion of the Order that is incorrect, obsolete, or otherwise in need of revision.
6. If, in the opinion of the Executive Officer, the Discharger fails to comply with the provisions of this Order, including compliance with the above limits, such failure to comply shall constitute adequate grounds to initiate action for administrative civil liability, pursuant to Section 13323, or to request the Attorney General take appropriate enforcement action pursuant to Section 13350 of the California Water Code.

1, Roger W. Briggs, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Coast Region, on December 3, 2004 and amended by Resolution No. R3-2005-0029 on February 11, 2005.



Roger W. Briggs

**STATE OF CALIFORNIA
CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL COAST REGION
895 Aerovista Place, SUITE 101
San Luis Obispo, California 93401**

**MONITORING AND REPORTING PROGRAM
NO. R3-2004-0153
Waste Discharger Identification No. 3 359803001
Revised December 3, 2004**

For

**SAN BENITO COUNTY
COUNTY SERVICES AREA NO. 45
RANCHO LARIOS
San Benito County**

A EFFLUENT MONITORING

Representative samples of treatment plant effluent shall be collected and analyzed for the following:

Constituent	Units	Sample Type	Sampling Frequency
Daily Flow	gal/day	Metered	Continuous
Turbidity	NTU	Grab	Continuous
Coliform	MPN/100 ml	Grab	Daily
Total Chlorine Residual	mg/L	"	"
Settleable Solids	ml/l	"	Daily
Total Dissolved Solids	mg/l	Grab	Monthly
Sodium	mg/l	"	"
Chloride	mg/l	"	"
BOD ₅	mg/l	"	"
pH	Units	"	"
Total Nitrogen	mg/l	"	"
Nitrate as N	mg/l	"	"

B WATER SUPPLY MONITORING

Representative samples of the water supply shall be collected and analyzed for the following:

Constituent	Units	Sample Type	Sampling Frequency
Total Dissolved Solids	mg/l	Grab	Monthly
Sodium	"	"	"
Chloride	"	"	"

C STORAGE FACILITY INSPECTION

Recycled water storage areas shall be inspected weekly. Notes shall be kept of observations and shall be summarized in annual monitoring reports. In the event of impending freeboard violation or storage pond overflow, the Executive Officer and the San Benito County Division of Environmental Health shall be notified immediately.

D RECYCLED WATER USE AREA MONITORING

1. During periods of Secondary-23 recycled water use, the Discharger shall inspect the recycled water use areas daily. Tertiary-2.2 recycled water use areas shall be inspected no less frequently than weekly. The status of compliance with these water recycling requirements shall be noted in a bound inspection log book. The logbook shall be made available to the Regional Board upon Executive Officer request. A summary of observations made during water recycling area inspections and a brief discussion of any corrective actions taken or planned shall be included with each quarterly monitoring report.
2. The Discharger shall cooperate with San Benito County Division of Environmental Health and California Department of Health Services to ensure that backflow devices are present, tested regularly, and repaired or replaced if found defective.
3. The Discharger shall review recycled water meter readings quarterly to identify any unusual usage changes. Any dramatic changes in usage shall be promptly investigated.

E REPORTING PROVISIONS

1. The Discharger shall immediately report any non-compliance potentially endangering public health or the environment to the Regional Board (805/549-3147) and the California Department of Health Services (805/566-1326), and any other appropriate agency. Any information shall be provided orally within 24 hours from the time the Discharger becomes aware of the circumstances. Such circumstances include:
 - a. Failure of chlorination equipment;
 - b. Tertiary treatment process effluent total coliform bacteria greater than 240 MPN/100 mL;
 - c. Tertiary treatment process effluent turbidity greater than 10 NTU; and
 - d. Tertiary treatment process disinfection process CT less than 450 mg-min/L.

A written report shall also be submitted to the Executive Officer within five (5) days of the time the Discharger becomes aware of the circumstances. The written report shall contain (1) a description of the non-compliance and its cause; (2) the period of non-compliance, including dates and times, and if the non-compliance has not been corrected, the anticipated time it is expected to continue; and (3) steps taken or planned to reduce, eliminate, and prevent reoccurrence of the non-compliance.

2. **Quarterly monitoring reports** shall be submitted within 30 days of the end of each quarter.

Monitoring Period	Report Submittal Date
January 1 – March 31	May 1 st
April 1 – June 30	August 1 st
July 1 – September 30	November 1 st
October 1 – December 31	February 1 st

3. **Annual monitoring reports** shall be submitted by January 30 of each year, in accordance with Standard Provisions and Monitoring Requirements. Annual reports shall evaluate trends and summarize the previous year's data.
4. **Salts Management Plan updates** - beginning with the 2006 annual report, salts management plan progress reports will be included in the annual report. These status reports will include all progress achieved to date in the control and reduction of the salts loading caused by the water reclamation and disposal activities.

5. Monitoring data shall be arranged in tabular format so the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in a manner that clearly illustrates compliance with water recycling requirements.
6. The Discharger shall deliver a copy of each monitoring report in the appropriate format to:

**California Regional Water Quality Control Board
Central Coast Region
895 Aerovista Place, Suite 101
San Luis Obispo, CA 93401**

F. SEWAGE SPILL REPORTING PROVISIONS

Reporting to the Regional Board

1. Sewage spills greater than 1,000 gallons and/or all sewage spills that enter a water body of the State, or occur where public contact is likely, regardless of the size, shall be reported to the Regional Board by telephone as soon as notification is possible and can be provided without substantially impeding cleanup or other emergency measures, and no later than 24 hours from the time that the Dischargers have knowledge of the overflow.
2. Unless fully contained, sewage spills to storm drains tributary to Waters of the United States shall be reported as discharges to surface waters.
3. A written report of all relevant information shall be submitted to the Regional Board within five days of the spill, and shall include no less information than is required on the current Sewage Spill Report Form (see Attachment 1), or equivalent, as approved by the Regional Board Executive Officer. Attachments to the report should be used as appropriate, and incidents requiring more time than the five-day period must be followed by periodic written status reports until issue closure. Photographs taken during the sewage spill incident and cleanup shall be submitted to the Regional Board in hard copy and electronic format.
4. The Dischargers shall sample all spills to surface waters to determine their effects on surface waters and submit the data to the Executive Officer in the next monthly monitoring report. Samples shall, at minimum, be analyzed for total and fecal coliform bacteria and enterococcus bacteria for spills to marine water, and fecal coliform bacteria for spills to fresh water. Sampling shall be conducted in the affected receiving water body upstream, at, and downstream of the spill's point of entry, and as necessary to characterize the spill's impact and to ensure adequate clean-up.
5. Spills under 1,000 gallons that do not enter a water body shall be reported to the Regional Board in writing and electronically (Excel spreadsheet preferred) within the next monthly monitoring report. Such reports shall include, at a minimum, a tabular summary of spill dates, locations, volumes, whether the spill discharged to surface waters (including conveyances thereto) or land, whether cleanup and/or disinfection was performed, the spill's cause, the number of spills at the location in the last three years, and weather conditions.

This policy is subject to revision by the Executive Officer.

6. The Dischargers shall submit to the Regional Board with the annual report required above, a summary of all spills between January 1 and December 31 of the previous year. The summary should include the following information for each spill:
 - a. Information requested in the Sewage Spill Report Form (Attachment 1);
 - b. How the spill volume was estimated and/or calculated;
 - c. Photograph(s) of spill, if taken;

- d. Where the spill entered any storm drain inlet or surface waters;
 - e. Steps taken or planned to reduce, eliminate, and prevent recurrence of the spill, and a schedule of major milestones for those steps;
 - f. Steps taken or planned to mitigate the impact(s) of the spill, and a schedule of major milestones for those steps;
 - g. Any additional correspondence and follow-up reports, as necessary, to supplement the Sewage Spill Report Form and to provide detailed information on cause, response, adverse effects, corrective actions, preventative measures, or other information.
7. The annual summary shall include detailed evaluations of repetitive or chronically occurring circumstances, such as problematic collection system areas or common spill causes, and the corrective actions taken to address such systematic problems.
 8. If no sewage spills occurred in the last calendar year, a statement certifying that no sewage spills occurred may be submitted in lieu of the annual summary.

Reporting to the Governor's Office of Emergency Services

9. In accordance with the Governor's Office of Emergency Services (OES) 2002 Fact Sheet regarding the reporting of sewage releases, the California Water Code, commencing with Section 13271, requires that a discharge of sewage to State waters must be reported to OES.
10. To report sewage releases of 1,000 gallons or more (currently the federal reportable quantity) to OES, verbally notify the OES Warning Center at: (800) 852-7550, or (916) 845-8911.
11. The following fax number should be used *for follow-up information only*: (916) 262-1677. The reportable quantity is subject to revision by the State of California. OES reporting requirements for sewage releases and hazardous materials can be located on the OES Website @ www.oes.ca.gov in the California Hazardous Material Spill/Release Notification Guidance. The OES Hazardous Materials Unit staff is available for questions at (916) 845-8741.
12. OES Reporting Exceptions: Notification to OES of an unauthorized discharge of sewage or hazardous substances is not required if: 1) the discharge to State waters is a result of a cleanup or emergency response by a public agency; 2) the discharge occurs on land only and does not affect State waters; or 3) the discharge is in compliance with applicable waste discharge requirements. These exceptions apply only to the Discharger's responsibility to report to OES, and do not alter the Regional Board's reporting policies or waste discharge requirements.

Ordered By: _____


Executive Officer

Date: _____

12-9-04

